Continuity & Differentiability

113. Let f : R $\to$ R be a continuous function satisfying f(0) = 1 and f(2x) – f(x) = x for all x $\in$ R. If $\lim_{n\to\infty}\left\{f(x)-f\left(\frac{x}{2^n}\right)\right\}=G(x)$, then $\sum_{r=1}^{10}G\left(r^2\right)$ is equal to :

माना एक संतत फलन f : R $\to$ R के लिए f(0) = 1 और f(2x) – f(x) = x, x $\in$ R हैं। यदि $\lim_{n\to\infty}\left\{f(x)-f\left(\frac{x}{2^n}\right)\right\}=G(x)$ है, तो $\sum_{r=1}^{10}G\left(r^2\right)$ बराबर है :

(A) 215 (B) 385 (C) 420 (D) 540

Ans. B

Sol. f : R $\to$ R

f(0) = 1

f(2x) – f(x) = x $\forall$ x $\in$ R

$$\underset{n\to\infty}{Lt}\left\{f(x)-f\left(\frac{x}{2^n}\right)\right\}=G(x)$$

then $\sum_{r=1}^{10}G\left(r^2\right)=??$

Now, f(2x) – f(x) = x

$$f(x)-f\left(\frac{x}{2}\right)=\frac{x}{2}$$

$$f\left(\frac{x}{2}\right)-f\left(\frac{x}{2^2}\right)=\frac{x}{2^2}$$

$$\vdots$$

$$f\left(\frac{x}{2^{n-1}}\right)-f\left(\frac{x}{2^n}\right)=\frac{x}{2^n}$$

$$f(2x)-f\left(\frac{x}{2^n}\right)=x\left\{1+\frac{1}{2}+\frac{1}{2^2}+.....\right\}$$

$n \to \infty$

$$f(2x)-f(0)=x-\frac{1}{1-\frac{1}{2}}\Rightarrow 2x$$

f(2x) = 2x + 1

$\therefore$ f(x) = x + 1

Now, $G(x) = f(x) - f(0)$

$= x + 1 - 1$

$= x$

Now, $\sum_{r=1}^{10} G\left(r^2\right) = \sum_{r=1}^{10} r^2$

$\Rightarrow 1^2 + 2^2 + ... + 10^2$

$\Rightarrow \frac{n(n+1)(2n+1)}{6} \quad (n = 10)$

$\Rightarrow \frac{10 \times 11 \times 21}{6}$

$\Rightarrow 5 \times 77$

$\Rightarrow 385$

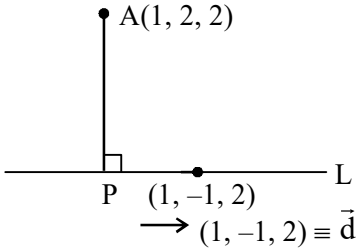
114. Let P be the foot of the perpendicular from point (1, 2, 2) on the line $L : \frac{x-1}{1} = \frac{y+1}{-1} = \frac{z-2}{2}$. Let the line $\vec{r} = \left(-\hat{i} + \hat{j} - 2\hat{k}\right) + \lambda\left(\hat{i} - \hat{j} + \hat{k}\right), \lambda \in R$, intersect the line L, at Q. Then $2(PQ)^2$ is equal to :

माना कि P, बिन्दु (1, 2, 2) से रेखा $L : \frac{x-1}{1} = \frac{y+1}{-1} = \frac{z-2}{2}$ पर लम्ब का पाद है। माना कि रेखा $\vec{r} = \left(-\hat{i} + \hat{j} - 2\hat{k}\right) + \lambda\left(\hat{i} - \hat{j} + \hat{k}\right), \lambda \in R$, रेखा L को Q पर काटती है। तब $2(PQ)^2$ बराबर है :

(A) 19 (B) 27 (C) 25 (D) 29

Ans. B

Sol.



$L : \frac{x-1}{1} = \frac{y+1}{-1} = \frac{z-2}{2} = \mu$

$= \mu P(\mu + 1, -\mu - 1, 2\mu + 2)$

$\overrightarrow{AP} \cdot \vec{d} = 0 \qquad \Rightarrow (\mu, -\mu - 3, 2\mu) \cdot (1, -1, 2) = 0$

$\Rightarrow \mu + \mu + 3 + 4\mu = 0 \qquad \Rightarrow \mu = -\frac{1}{2}$

$\therefore P\left(\frac{-1}{2} + 1, +\frac{1}{2} - 1, 2\left(\frac{-1}{2}\right) + 2\right)$

$P\left(\frac{1}{2},\frac{-1}{2},1\right)$

Now general pt. on L_2 is $Q(-1+\lambda, 1-\lambda, -2+\lambda)$

Equate it with general pt of L

$\mu + 1 = -1 + \lambda$ | $-\mu - 1 = 1 - \lambda$ | $2\mu + 2 = -2 + \lambda$

$\mu = \lambda - 2$ | $\mu = \lambda - 2$ | $\downarrow$

$2(\lambda - 2) + 2 = -2 + \lambda$

$2\lambda - 4 + 2 = -2 + \lambda$

$\therefore \mu = -2, \lambda = 0$

$\therefore Q \equiv -2, \lambda = 0$

$P\left(\frac{1}{2},\frac{-1}{2},1\right)$ and $Q(-1,1,-2)$

$$PQ = \sqrt{\left(\frac{1}{2}+1\right)^2 + \left(\frac{-1}{2}-1\right)^2 + (1+2)^2}$$

$$= \sqrt{\frac{9}{4}+\frac{9}{4}+9} = \sqrt{\frac{54}{4}}$$

$$\therefore 2(PQ)^2 = 2\left(\frac{54}{4}\right) = 27$$

115. Let a straight line L pass through the point P (2, –1, 3) and be perpendicular to the lines $\frac{x-1}{2}=\frac{y+1}{1}=\frac{z-3}{-2}$ and $\frac{x-3}{1}=\frac{y-2}{3}=\frac{z+2}{4}$. If the line L intersects the yz-plane at the point Q, then the distance between the points P and Q is :

माना कि एक सरल रेखा L बिन्दु P (2, –1, 3) से होकर जाती है और और रेखाओं $\frac{x-1}{2}=\frac{y+1}{1}=\frac{z-3}{-2}$ और $\frac{x-3}{1}=\frac{y-2}{3}=\frac{z+2}{4}$ के लम्बवत् है। यदि रेखा L, yz- तल को बिन्दु Q पर काटती है, तो बिन्दुओं P और Q के बीच की दूरी है :

(A) $2\sqrt{3}$ (B) 2 (C) 3 (D) $\sqrt{10}$

Ans. C

Sol. Vector parallel to 'L'

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -2 \\ 1 & 3 & 4 \end{vmatrix} = 10\hat{i} - 10\hat{j} + 5\hat{k}$$

$$= 5\left(2\hat{i} - 2\hat{j} + \hat{k}\right)$$

Equatino of 'L' $\frac{x-2}{2}=\frac{y+1}{-2}=\frac{z-3}{1}=\lambda$ (say)

Let $Q(2\lambda+2, -2\lambda-1, \lambda+3)$

$\Rightarrow 2\lambda+2=0 \qquad \Rightarrow \lambda=-1$

$\Rightarrow Q(0, 1, 2)$

$d(P, Q) = 3$

116. Let α, β $(\alpha \neq \beta)$ be the values of m, for which the equations $x + y + z = 1$; $x + 2y + 4z = m$ and $x+4y+10z=m^2$ have infinitely many solutions. Then the value of $\sum_{n=1}^{10}\left(n^{\alpha}+n^{\beta}\right)$ is equal to :

माना कि α, β $(\alpha \neq \beta)$, m के वे मान है, जिनके लिए समीकरणों $x + y + z = 1$; $x + 2y + 4z = m$ और $x + 4y + 10z = m^2$ के अनन्त हल हैं। तब $\sum_{n=1}^{10}\left(n^{\alpha}+n^{\beta}\right)$ का मान बराबर है :

(A) 440 (B) 3080 (C) 3410 (D) 560

Ans. A

Sol. $\Delta=\begin{vmatrix}1 & 1 & 1\\ 1 & 2 & 4\\ 1 & 4 & 10\end{vmatrix}=1(20-16)-1(10-4)+1(4-2)$

$= 4 - 6 + 2 = 0$

For infinite solutions

$\Delta_x = \Delta_y = \Delta_z = 0$

$m^2 - 3x + 2 = 0$

$m = 1, 2$

$\alpha = 1, \beta = 2$

$\therefore \sum_{n=1}^{10}\left(n^{\alpha}+n^{\beta}\right)=\sum_{n=1}^{10}n^1+\sum_{n=1}^{10}n^2$

$=\frac{10(11)}{2}+\frac{10(11)(21)}{6}$

$= 55 + 385$

$= 440$

117. Let $f : [-1, 2] \to R$ be given by $f(x) = 2x^2 + x + [x^2] - [x]$, where [t] denotes the greatest integer less than or equal to t. The number of points, where f is not continuous is :

माना फलन $f : [-1, 2] \to R$, $f(x) = 2x^2 + x + [x^2] - [x]$ द्वारा दिया गया है, जहाँ [t] महत्तम पूर्णांक $\le t$ हैं। उन बिन्दुओं जिन पर f संतत नहीं है, की संख्या है) :

(A) 3 (B) 5 (C) 6 (D) 4

Ans. D

Sol. Doubtful points : $-1, 0, 1, \sqrt{2}, \sqrt{3}, 2$

at $x = \sqrt{2}, \sqrt{3}$

$f(x) = (2x^2 + x - [x]) + [x^2]$ =Discontinuous

$\downarrow$ $\downarrow$

Cont. Discont.

at $x = -1$:

$$\left.\begin{aligned} \text{RHL} \Rightarrow f(x) &= (2 - 1 - (-1)) + 0 = 2 \\ F(-1) &= 2 - 1 - (-1) + 1 = 3 \end{aligned}\right\} \text{Dis.}$$

at $x = 2$:

$$\left.\begin{aligned} \text{LHL} \Rightarrow f(x) 8 + 2 - 1 + 3 &= 12 \\ F(2) &= 8 + 2 - 2 + 4 = 12 \end{aligned}\right\} \text{Cont.}$$

at $x = 0$:

$$\left.\begin{aligned} \text{LHL} \Rightarrow 0 + 0 - (-1) + 0 &= 1 \\ F(0) &= 0 \end{aligned}\right\} \text{Dis.}$$

at $x = 1$:

$$\left.\begin{aligned} \text{LHL} \Rightarrow 2 + 1 - 0 + 0 &= 3 \\ F(1) &= 3 - 1 + 1 = 3 \\ \text{RHL} \Rightarrow 2 + 1 - 1 + 1 &= 3 \end{aligned}\right\} \text{Cont.}$$

118. For $a, b > 0$, let $f(x) = \begin{cases} \dfrac{\tan((a+1)x) + b\tan x}{x}, & x < 0 \\ 3, & x = 0 \\ \dfrac{\sqrt{ax + b^2x^2} - \sqrt{ax}}{b\sqrt{a}x\sqrt{x}}, & x > 0 \end{cases}$ be a continuous function at $x = 0$. Then $\dfrac{b}{a}$ is equal to

:

$a, b > 0$ के लिए, माना $f(x) = \begin{cases} \dfrac{\tan((a+1)x) + b\tan x}{x}, & x < 0 \\ 3, & x = 0 \\ \dfrac{\sqrt{ax + b^2x^2} - \sqrt{ax}}{b\sqrt{a}x\sqrt{x}}, & x > 0 \end{cases}$, $x = 0$, $x = 0$ पर एक संतत फलन है। तो $\dfrac{b}{a}$ बराबर है :

(A) 5 (B) 6 (C) 4 (D) 8

Ans. B

Sol. $\lim\limits_{x\to 0} f(x) = f(0) = 3$

$$= \lim_{x\to 0^+} \frac{\sqrt{ax + b^2x^2} - \sqrt{ax}}{b\sqrt{a}x\sqrt{x}} = 3$$

$$= \lim_{x\to 0^+} \frac{ax + b^2x^2 - ax}{b\sqrt{a}x^{\frac{3}{2}}\left(\sqrt{ax + b^2x^2} + \sqrt{ax}\right)}$$

$$= \lim_{x\to 0^+} \frac{b^2}{b\sqrt{a}\left(\sqrt{a + b^2x} + \sqrt{a}\right)}$$

$$= \frac{b}{\sqrt{a}\cdot 2\sqrt{a}} \Rightarrow \frac{b}{2a} = 3 \qquad ...(1)$$

$$= \lim_{x\to 0^-} \frac{\tan(a+1)x + b\tan x}{x}$$

$$= \lim_{x\to 0^-} \frac{\dfrac{\tan(a+1)x}{(a+1)x}(a+1)x + \dfrac{b\tan x}{x}\cdot x}{x}$$

$= a + 1 + b = 3 \Rightarrow a + b = 2$ ...(2)

from (1) and (2)

$a + 6a = 2$

$a = \dfrac{2}{7}$ $b = \dfrac{12}{7}$

119. Consider the function.

$$f(x)=\begin{cases}\dfrac{a\left(7x-12-x^2\right)}{b\left|x^2-7x+12\right|} & , \quad x<3\\ 2^{\frac{\sin(x-3)}{x-[x]}} & , \quad x>3\\ b & , \quad x=3\end{cases}$$

where [x] denotes the greatest integer less than or equal to x. If S denotes the set of all ordered pairs (a, b) such that f(x) is continuous at x = 3, then the number of elements in S is :

(A) 1 (B) 2 (C) Infinitely many (D) 4

फलन

$$f(x)=\begin{cases}\dfrac{a\left(7x-12-x^2\right)}{b\left|x^2-7x+12\right|} & , \quad x<3\\ 2^{\frac{\sin(x-3)}{x-[x]}} & , \quad x>3\\ b & , \quad x=3\end{cases}$$

जहाँ [x] महत्तम पूर्णांक फलन है, का विचार कीजिए। यदि सभी क्रमित युग्मों (a, b), जिनके लिए x = 3 पर f(x) संतत है, का समुच्चय S है, तो में अवयवों की संख्या है :

(A) 1 (B) 2 (C) अनन्त (D) 4

Ans. A

Sol. $f\left(3^-\right)=\dfrac{a}{b}\dfrac{\left(7x-12-x^2\right)}{\left|x^2-7x+12\right|}$ (for f(x) to be cont.)

$$\Rightarrow f\left(3^-\right)=\frac{-a}{b}\frac{(x-3)(x-4)}{(x-3)(x-4)};x<3\Rightarrow\frac{-a}{b}$$

Hence $f\left(3^{-}\right)=\frac{-a}{b}$

Then $f\left(3^{+}\right)=2^{\lim\limits_{x\to3^{+}}\left(\frac{\sin(x-3)}{x-3}\right)}=2$ and

f(3) = b.

Hence $f(3)=f\left(3^{+}\right)=f\left(3^{-}\right)$

$\Rightarrow b=2=-\frac{a}{b}$

b = 2, a = –4

Hence only 1 ordered pair (–4, 2)

120. Consider the function f : (0, 2) → R defined by $f(x)=\frac{x}{2}+\frac{2}{x}$ and the function g(x) defined by

$$g(x)=\begin{cases}\min(f(t)), & 0<t\le x \text{ and } 0<x\le 1\\ \frac{3}{2}+x, & 1<x<2\end{cases}$$. Then :

(A) g is continuous but not differentiable at x=1 (B) g is continuous and differentiable for all x ∈ (0, 2)

(C) g is not continuous for all x ∈ (0, 2) (D) g is neither continuous nor differentiable at x = 1

$f(x)=\frac{x}{2}+\frac{2}{x}$ द्वारा परिभाषित फलन f : (0, 2) → R तथा $g(x)=\begin{cases}\min(f(t)), & 0<t\le x \text{ and } 0<x\le 1\\ \frac{3}{2}+x, & 1<x<2\end{cases}$ द्वारा परिभाषित

फलन g(x) का विचार कीजिए। तो :

(A) x = 1 पर g संतत है परन्तु अवकलनीय नहीं है। (B) सभी x ∈ (0, 2) के लिए g संतत तथा अवकलनीय है।

(C) सभी x ∈ (0, 2) के लिए g संतत नहीं है। (D) x = 1 पर g न तो संतत है न ही अवकलनीय है।

Ans. A

Sol. $f(x)=\frac{x}{2}+\frac{2}{x}$

$\Rightarrow f'(x)=\frac{1}{2}-\frac{2}{x^2}=\left(\frac{x^2-4}{2x^2}\right)$ for x ∈ (0, 2)

$\Rightarrow f(x)<0$ $\Rightarrow$ f(x) is decreasing

$$g(x)=\begin{cases}\min\{f(t)\} & ,0<t\le x \text{ and } 0<x\le 1\\ \frac{3}{2}+x & ,1<x<2\end{cases}$$

Since f(x) is decreasing, f(t) is minimum for x in $t\in(0,x]\Rightarrow \min\{f(t)\}=f(x)$

$$\Rightarrow g(x)=\begin{cases}f(x) & , \quad x\in(0,1]\\ \frac{3}{2}+x & , \quad 1<x<2\end{cases}$$

$$\Rightarrow g(1)=\frac{1}{2}+\frac{2}{1}=\frac{5}{2}$$

Also $g(1^+)=\frac{3}{2}+1=\frac{5}{2}$

$\Rightarrow$ g(x) is continuous in (0, 2)

$$g'(x)=\begin{cases}f'(x) & , \quad x\in(0,1]\\ 1 & , \quad x\in(1,2)\end{cases}$$

$$g'(1)=f'(1)=\frac{1-4}{2}=\frac{-3}{2}$$

$\Rightarrow g'(1)\neq g'(1^+)\Rightarrow$ not differentiable at x = 1

121. Let $f:R-\{0\}\to R$ be a function satisfying $f\left(\frac{x}{y}\right)=\frac{f(x)}{f(y)}$ for all $x,y,f(y)\neq 0$. If $f'(1)=2024$, then :

माना एक फलन $f:R-\{0\}\to R$ सभी $x,y,f(y)\neq 0$ के लिए $f\left(\frac{x}{y}\right)=\frac{f(x)}{f(y)}$ को संतुष्ट करता है। यदि $f'(1)=2024$ है, तो :

(A) $xf'(x)+f(x)=2024$

(B) $xf'(x)-2024f(x)=0$

(C) $xf'(x)+2024f(x)=0$

(D) $xf'(x)-2023f(x)=0$

Ans. B

Sol. $f\left(\frac{x}{y}\right)=\frac{f(x)}{f(y)}$

$f'(1)=2024$

$f(1)=1$

Partially differentiating w.r.t. x

$$f'\left(\frac{x}{y}\right)\cdot\frac{1}{y}=\frac{1}{f(y)}f'(x)$$

$y \to x$

$$f'(1)\cdot\frac{1}{x}=\frac{f'(x)}{f(x)}$$

$2024f(x) = xf'(x) \Rightarrow xf'(x) - 2024f(x) = 0$

122. Consider the function $f:(0,\infty)\to R$ defined by $f(x)=e^{-|\log_e x|}$. If m and n be respectively the number of points at which f is not continuous and f is not differentiable, then m + n is :

$f(x)=e^{-|\log_e x|}$ द्वारा परिभाषित फलन $f:(0,\infty)\to R$ का विचार कीजिए। यदि उन बिंदुओं की संख्या, जहाँ f संतत नहीं है तथा अवकलनीय नहीं हैं क्रमशः m तथा n है, तो m + n बराबर है :

(A) 2 (B) 3 (C) 0 (D) 1

Ans. D

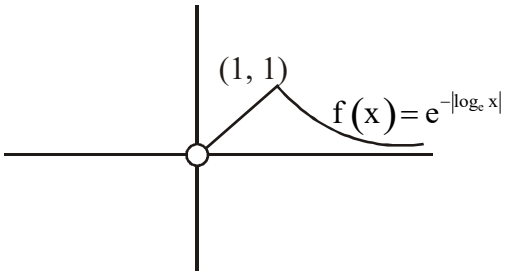
Sol. $f(x)=e^{-|\ln x|}$

$f:(0,\infty)\to R$

$f(x)=e^{-|\log_e x|}$

$$f(x)=\frac{1}{e^{|\ln x|}}=\begin{cases}\frac{1}{e^{-\ln x}}, & 0<x<1\\ \frac{1}{e^{\ln x}}, & x\ge 1\end{cases}$$

$$=\begin{cases}x, & 0<x<1\\ \frac{1}{x}, & x\ge 1\end{cases}$$



f(x) is continuous in $(0,\infty) \Rightarrow m = 0$
from graph, f(x) is not differentiable at 1 point at x = 1
$\Rightarrow n = 1$

therefore m + n = 1

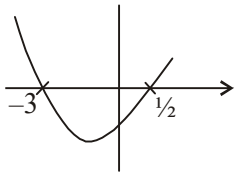
123. Let $f(x)=\left|2x^2+5|x|-3\right|, x\in R$. If m and n denote the number of points where f is not continuous and not differentiable respectively, then m + n is equal to :

माना $f(x)=\left|2x^2+5|x|-3\right|, x \in R$ हैं। यदि m तथा n क्रमशः उन बिंदुओं, जहाँ पर f संतत नहीं है तथा अवकलनीय नहीं है की संख्या हैं, तो m + n बराबर है :

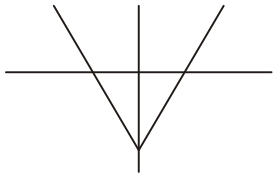
(A) 5 (B) 0 (C) 3 (D) 2

Ans. C

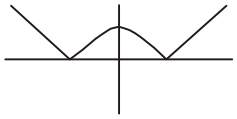
Sol. $y = 2x^2 + 5x - 3$



$y = 2x^2 + 5|x| - 3$



$y = |2x^2 + 5|x| - 3|$



$m = 0, n = 3$

$\Rightarrow m + n = 3$

124. Let m and n be the number of points at which the function $f(x) = \max\{x, x^3, x^5, ..., x^{21}\}, x \in R$, is not differentiable and not continuous, respectively. Then m + n is equal to _____.

माना उन बिन्दुओं, जहाँ पर फलन $f(x) = \max\{x, x^3, x^5, ..., x^{21}\}, x \in R$, अवकलनीय नहीं है और संतत नहीं है, की संख्या क्रमशः m और n है। तो m + n बराबर है _____.

Ans. 3

Sol. $f(x) = \max\{x, x^3, x^5, ... x^{21}\}$

$$f(x)=\begin{cases} x & ; \quad x<-1 \\ x^{21} & ; \quad -1 \le x<0 \\ x & ; \quad 0 \le x<1 \\ x^{21} & ; \quad x \ge 1 \end{cases}$$

f(x) is continuous everywhere

n = 0

$$f'(x)=\begin{cases}1 & ; & x<-1\\ 21x^{20} & ; & -1\le x<0\\ 1 & ; & 0<x<1\\ 21x^{20} & ; & x\ge 1\end{cases}$$

f(x) is non-differentiable at x = –1, 0, 1

$\therefore$ m = 3

m + n = 3

125. The number of points of discontinuity of the function $f(x)=\left[\frac{x^2}{2}\right]-\left[\sqrt{x}\right]$, $x\in[0,4]$, where [•] denotes the greatest integer function, is _____.

फलन $f(x)=\left[\frac{x^2}{2}\right]-\left[\sqrt{x}\right]$, $x\in[0,4]$, जहाँ where [•] महत्तम पूर्णांक फलन है, के असांतत्य के बिन्दुओं की संख्या है _____.

Ans. 8

Sol. $x=0,1,\sqrt{2},\sqrt{4},\sqrt{6},\sqrt{8},\sqrt{10},\sqrt{12},\sqrt{14},4$

$0\le x^2\le 16$

$0\le\frac{x^2}{2}\le 8$

$x^2=0,2,4,6,8,10,12,14,16$

$x=0,\sqrt{2},2,\sqrt{6},2\sqrt{2},\sqrt{10},2\sqrt{3},\sqrt{14},4$

$\frac{x^2}{2}=0,1,2,3,4,5,6,7,8$ $\rightarrow$ 8 points

$0\le\sqrt{x}\le 2$

$\sqrt{x}=0,1,2$ $\rightarrow$ 2 points

x = 0,1,4

f(4) = 8 – 2 = 6

$f(4^-)$ = 7 – 1 = 6

Total points = 8 + 2 – 2 = 8 points

126. If the function $f(x)=\frac{\tan(\tan x)-\sin(\sin x)}{\tan x-\sin x}$ is continuous at x = 0, then f(0) is equal to _____.

यदि फलन $f(x)=\frac{\tan(\tan x)-\sin(\sin x)}{\tan x-\sin x}$, x = 0 पर संतत है, तो f(0) बराबर है _____.

Ans. 2

Sol. $f(x)=\lim\limits_{x\to 0}\dfrac{\tan(\tan x)-\sin(\sin x)}{\tan x-\sin x}$

by expansion

$$=\lim_{x\to 0}\frac{\left(\tan x+\frac{(\tan x)^3}{3}+....\right)-\left((\sin x)-\frac{(\sin x)^3}{3!}\right)}{\left(x+\frac{x^3}{3}-\left(x-\frac{x^3}{3!}\right)\right)}$$

$$=\lim_{x\to 0}\frac{\left(x+\frac{x^3}{3}\right)+\frac{1}{3}\left(x+\frac{x^3}{3}\right)^3-\left(x-\frac{x^3}{3!}\right)+\frac{1}{6}\cdot\left(x-\frac{x^3}{6}\right)^3}{\frac{x^3}{2}}$$

$$=\lim_{x\to 0}\frac{x^3\left[\frac{1}{3}+\frac{1}{3}+\frac{1}{6}+\frac{1}{6}\right]}{\frac{x^3}{2}}=2$$

127. Let $f(x)=\begin{cases}3x, & x<0\\ \min\{1+x+[x],x+2[x]\}, & 0\le x\le 2\\ 5, & x>2,\end{cases}$

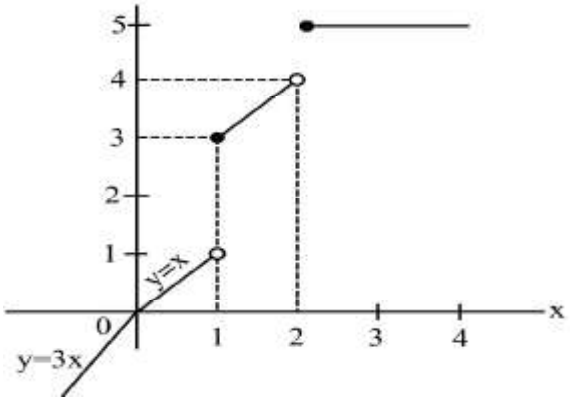
where [.] denotes greatest integer function. If α and β are the number of points, where f is not continuous and is not differentiable, respectively, then $\alpha+\beta$ equals _____.

माना कि $f(x)=\begin{cases}3x, & x<0\\ \min\{1+x+[x],x+2[x]\}, & 0\le x\le 2\\ 5, & x>2,\end{cases}$ है,

जहाँ [.] महत्तम पूर्णांक फलन को निर्दिष्ट करता है। यदि α और β क्रमशः उन बिन्दुओं की संख्याएँ हैं, जहाँ f संतत् नहीं है और अवकलनीय नहीं है, तो $\alpha+\beta$ बराबर है _____.

Ans. 5

Sol. $f(x)=\begin{cases}3x & ; & x<0\\ x & ; & 0\le x<1\\ x+2 & ; & 1\le x<2\\ 5 & ; & x>2\end{cases}$



Not continuous at $x \in \{1, 2\} \Rightarrow \alpha = 2$
Not differentiable at $x \in \{0,1,2\} \Rightarrow \beta = 3$
$\alpha + \beta = 5$

128. Let [t] denote the greatest integer less than or equal to t. Let $f : [0, \infty) \to R$ be a function defined by $f(x) = \left[\frac{x}{2} + 3\right] - \left[\sqrt{x}\right]$. Let S be the set of all points in the interval [0, 8] at which f is not continuous. Then $\sum_{a \in S} a$ is equal to _____.

माना [t], t से कम या t के बराबर महत्तम पूर्णांक को दर्शाता है। माना फलन $f : [0, \infty) \to R$, $f(x) = \left[\frac{x}{2} + 3\right] - \left[\sqrt{x}\right]$ द्वारा परिभाषित है। माना S, अन्तराल [0, 8] में उन सभी बिन्दुओं का समुच्चय है, जहाँ f संतत नहीं है। तो $\sum_{a \in S} a$ बराबर है _____

Ans. 17

Sol. $f(x) = \left[\frac{x}{2} + 3\right] - \left[\sqrt{x}\right]$

$= \left[\frac{x}{2}\right] - \left[\sqrt{x}\right] + 3 \qquad \because \in [0,8], \frac{x}{2} \in [0,4], \sqrt{x} \in \left[0, 2\sqrt{2}\right]$

Point of discontinuity

When $\frac{x}{2}$ is integer & $\sqrt{x}$ is int.

$\frac{x}{2} = 0, 1, 2, 3, 4 \qquad \sqrt{x} = 0, 1, 4, 9$

$x = \underset{\substack{\downarrow \\ \text{Right} \\ \text{cont}}}{0}, \underbrace{2,4,6,8}_{\text{discont}} \qquad x = \underset{\substack{\downarrow \\ \text{cont}}}{0}, \underbrace{1,4,9}_{\text{discont}}$

$\therefore$ f(x) is discontinuous at x = 1,2,6,8

Check continuity at x = 4

$$f\left(4^+\right)=\left[2^+\right]-\left[\sqrt{4^+}\right]+3=3$$

$$f\left(4^-\right)=\left[2^-\right]-\left[\sqrt{4^-}\right]+3=1-1+3=3$$

$$f(4)=[2]-\left[\sqrt{4}\right]+3=3$$

$\therefore$ f(x) is continuous at x = 4

Point of discontinuity = 1, 2, 6, 8

$$\sum_{a\in s} a=1+2+6+8=17$$

129. Let $f:(0,\pi)\to R$ be a function given by $f(x)=\begin{cases}\left(\frac{8}{7}\right)^{\frac{\tan 8x}{\tan 7x}}, & 0<x<\frac{\pi}{2}\\ a-8, & x=\frac{\pi}{2}\\ \left(1+|\cot x|\right)^{\frac{b}{a}|\tan x|}, & \frac{\pi}{2}<x<\pi\end{cases}$ where a, b $\in$ Z. If f is continuous at $x=\frac{\pi}{2}$, then $a^2 + b^2$ is equal to_____.

माना $f:(0,\pi)\to R$ एक फलन दिया गया है जो कि $f(x)=\begin{cases}\left(\frac{8}{7}\right)^{\frac{\tan 8x}{\tan 7x}}, & 0<x<\frac{\pi}{2}\\ a-8, & x=\frac{\pi}{2}\\ \left(1+|\cot x|\right)^{\frac{b}{a}|\tan x|}, & \frac{\pi}{2}<x<\pi\end{cases}$ है, जहाँ a, b $\in$ Z है। यदि $x=\frac{\pi}{2}$ पर संतत है, तो $a^2 + b^2$ is बराबर है_____

Ans. 81

Sol. As f(x) is continuous at x

$$\Rightarrow Lf\left(\frac{\pi}{2}\right)=f\left(\frac{\pi}{2}\right)=Rf\left(\frac{\pi}{2}\right)$$

$$\Rightarrow \lim_{x\to\frac{\pi}{2}}\left(\frac{8}{7}\right)^{\frac{\tan 8x}{\tan 7x}} = a-8 = \lim_{x\to\frac{\pi}{2}}\left(1+|\cot x|\right)^{\frac{b}{a}|\tan x|}$$

$$\Rightarrow \left(\frac{8}{7}\right)^{0} = a-8 = e^{\frac{b}{a}}$$

$\Rightarrow a = 9$ and $b = 0$

Hence $a^2 + b^2 = 81$

130. Let for a differentiable function $f:(0,\infty)\to R$, $f(x)-f(y)\geq \log_e\left(\frac{x}{y}\right)+x-y$, $\forall\, x,y\in(0,\infty)$. Then $\sum_{n=1}^{20} f'\left(\frac{1}{n^2}\right)$ is equal to _____.

माना एक अवकलनीय् फलन $f:(0,\infty)\to R$ के लिए, $f(x)-f(y)\geq \log_e\left(\frac{x}{y}\right)+x-y$, $\forall\, x,y\in(0,\infty)$ है। तो $\sum_{n=1}^{20} f'\left(\frac{1}{n^2}\right)$ बराबर है _____।

Ans. 2890

Sol. $f(x)-f(y)\geq \ln x-\ln y+x-y$

$$\frac{f(x)-f(y)}{x-y}\geq \frac{\ln x-\ln y}{x-y}+1$$

Let $x>y$

$$\lim_{y\to x} f'(x^-)\geq \frac{1}{x}+1 \qquad \text{_____(1)}$$

Let $x<y$

$$\lim_{y\to x} f'(x^+)\leq \frac{1}{x}+1 \qquad \text{_____(2)}$$

$$f^1(x^-) = f^1(x^+)$$

$$f^1(x) = \frac{1}{x}+1$$

$$f'\left(\frac{1}{x^2}\right) = x^2+1$$

$$\sum_{x=1}^{20}\left(x^2+1\right) = \sum_{x-1}^{20} x^2+20$$

$$= \frac{20\times 21\times 41}{6}+20$$

$= 2890$